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Date ____ / ____ / ____

Bitwise Operators, For loop,

Variable Scope, Operator precedence

Bitwise operators

- AND (&)
- OR (|)
- XOR (^)
- NOT (~)

example. $a=2$ $b=3$ then $a \otimes b$

$$2 = 10$$
$$3 = 11$$

8

1	0
---	---

 $\rightarrow$ 2

And table

OR Table

Not table

X	Y	$X \& Y$
0	0	0
0	1	0
1	0	0
1	1	1

X	Y	X Y
0	0	0
0	1	1
1	0	1
1	1	1

X	$\sim X$
0	1
1	0

$$a = 2 \equiv 10$$
$$a=2 \equiv 10$$
$$b \equiv 9 \equiv 100$$
$$6 \equiv 110$$

$a = 00000000 \dots 10$
32 bit = 4 byte

XOR Table

x	y	$x \wedge y$
0	0	0
0	1	1
1	0	1
1	1	0

$$a = 2 \equiv 016$$
$$b_2 \neq 100$$
$$a^b = 119$$

$\sim a = 00011 = 3$
 $\sim a = -3$

$\sim q_2 = \textcircled{1} 11111 \dots 01$

NE apply 1's then 2's

$9 \cdot 0 \cdot 0 \cdot 0 \cdot 0 \cdot 0 \cdot 0 \cdot 0$

00000000000011

$$= -3$$

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left and right shift operators.

left shift: $5 << 1$

→ 5 को एक बार left shift

$5 \equiv 101$

$5 << 1 \equiv 1010 \equiv 10$

for smaller number only

Right shift: similar as the left shift

$5 >> 2$

$5 \equiv 101$

$5 >> 2 \equiv 1 \equiv 1$

trick
for left shift = $\times 2$
for right shift = $\div 2$

Right and left shift for +ve number then padding with zero. for negative padding is compiler dependent.

Increment Decrement Operator

$i++;$ → post increment (use करें के बाद i बढ़ जायगा)

$++i;$ → pre-increment (पहले increment)

$i--;$ → post decrement

$--i;$ → pre decrement

int i = 4

int a = i++;

printf(a)

printf(i)

output 4
5

int r = 11

int a = ++i

printf(a)

printf(i)

output: 12
12

Similar is apply on decrement operator

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For loop:

```
for ( ____ : ____ : ____ ) {
```

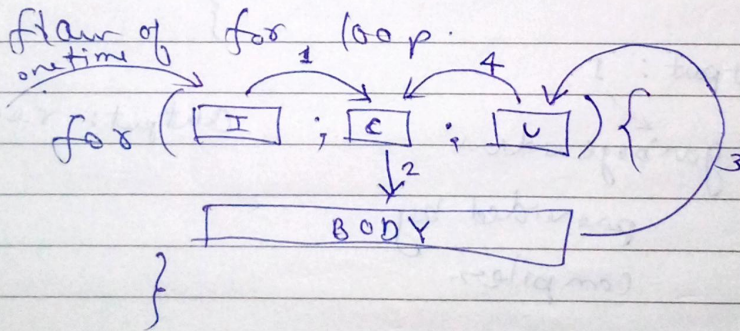
↓ initialization
 ↓ condition
 ↓ updation

Here all three are optional

we can also write two initializ, two conditions and two updation also.

example

```
for (int a=0, b=1; a<=0 && b>=1; a--, b--) {
    cout << a << " " << b << endl;
}
```



I = initialization

C = condition

U = updation

continue: skip a particular iteration.

break: iteration se hi bahar

example: of continue statement

```
for (int i = 0; i < 5; i++)
{
    cout << "Hi";
    cout << "Hey";
    continue;
    cout << "reply to kar de";
}
```

Variables and their Scope:

```
int main() {
    int a;
    cout << a << endl;
}
```

output: 1

garbage value
provided by
compiler.

```
int main() {
    int b = 2;
    int b = 3;
    cout << b << endl;
}
```

output: redefinition of b

```
int main() {
    int i = 8;
    for (int i = 0; i < 8; i++) {
        cout << "Hi" << endl;
    }
}
```

output: 8 times Hi from 0 to 7

Precedence And Associativity of OPERATORS

OPERATOR	DESCRIPTION	ASSOCIATIVITY
()	parenthesis	left to right
**	Exponent	right to left
// * / %	Multiplication/division/modules	left to right
+ -	Addition/subtraction	left to right
<< >>	Bitwise shift left, Bitwise shift right	left to right
<<=	Relational less than/less than or equal to	left to right
>>=	Relational greater than/greater than or equal to	left to right
= !=	Relational is equal to/is not equal to	left to right
is, is not in, not in	identity membership operators	left to right
&	Bitwise AND	left to right
^	Bitwise exclusive OR	left to right
!	Bitwise inclusive OR	left to right
not	logical NOT	right to left
and	logical AND	left to right
or	logical OR	left to right
=	Assignment	right to left
+= -=	Addition/subtraction assignment	
*= /=	Multiplication/division assignment	
%= &=	Modulus/bitwise AND assignment	
^= =	Bitwise exclusive/inclusive OR assignment	
<<= >>=	Bitwise shift left/right assignment	